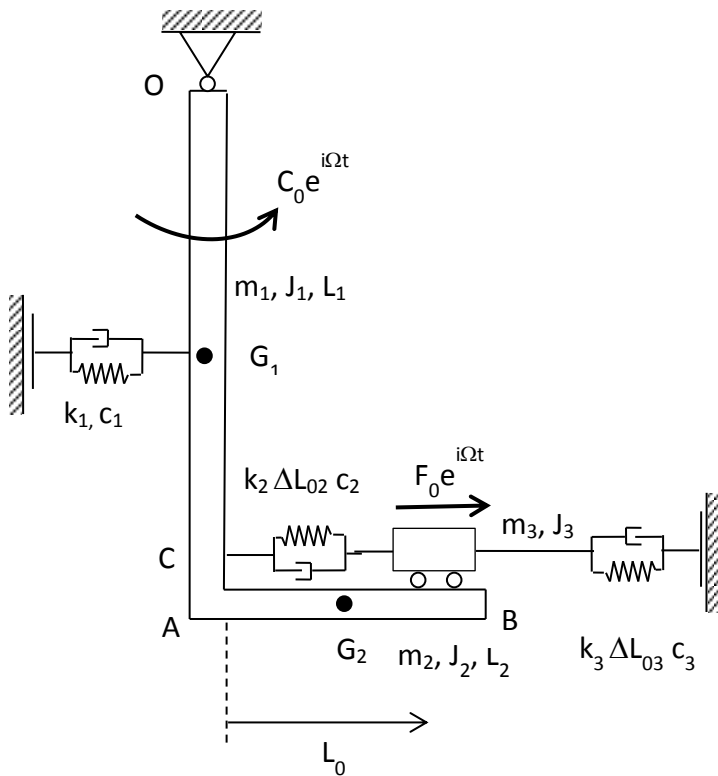


Mechanical Systems Dynamic

December 11th, 2014 – Version B



$m_1 = 10 \text{ kg}$	$k_1 = 3000 \text{ N/m}$	$F_{01} = 100 \text{ N}$
$m_2 = 5 \text{ kg}$	$k_2 = 1000 \text{ N/m}$	$C_{02} = 10 \text{ Nm}$
$m_3 = 3 \text{ kg}$	$k_3 = 1000 \text{ N/m}$	$\varphi_1 = \pi/6 \text{ rad}$
$J_1 = 0.2 \text{ kgm}^2$	$\Delta L_{02} = 0.02 \text{ m}$	$\varphi_2 = \pi/3 \text{ rad}$
$J_2 = 0.02 \text{ kgm}^2$	$\Delta L_{03} = 0.02 \text{ m}$	$\Omega_1 = 2\pi \text{ rad/s}$
$J_3 = 0.01 \text{ kgm}^2$	$c_1 = 10 \text{ Ns/m}$	$\Omega_2 = 4\pi \text{ rad/s}$
$L_1 = 0.50 \text{ m}$	$c_2 = 0.5 \text{ Ns/m}$	
$L_2 = 0.25 \text{ m}$	$c_3 = 0.5 \text{ Ns/m}$	
$L_0 = 0.10 \text{ m}$		

The mechanical system above lays in the vertical plane and is shown in its equilibrium position. It is constituted by a “L shaped” rigid body and by a second rigid body (m_3, J_3), which is sliding on the horizontal side of the first body (consider negligible the distance AC). The rigid bodies are connected each other through a spring and damper (k_2, c_2). The static preload of the springs k_2 and k_3 is given and is equal to ΔL_{02} and ΔL_{03} respectively. In the static configuration the distance between the center of gravity of the mass m_3 and point C is L_0 .

The students are requested to:

1. Write the linearized equations of motion around the given static configuration;
2. Compute the natural frequencies and modes of vibrations;
3. Compute the Frequency Response Function (in the range 0 – 5 Hz, step = 0.01 Hz) applying an external horizontal force at mass m_3 . Output: the rotation of the “L shaped” rigid body and the force transmitted by the spring and damper system (k_3, c_3). Save the graphical results as SNxxx1.fig and SNxxx2.fig, whit S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your identification number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
4. Compute the static preload of the k_1 spring (write also the numerical result on the sheets of the solution);
5. Compute the time history of the vertical component of the constraint force of the hinge O, given the following inputs: a horizontal force $F_{01}e^{i(\Omega_1 t + \varphi_1)}$ applied to the mass m_3 , and a torque $C_{02}e^{i(\Omega_2 t + \varphi_2)}$ applied to the “L shaped” rigid body. Save the results as SNxxx3.fig.

Note: Please collect the matlab script file relative to the numerical solution of the exercise and save it with the name SNxxx.m, where S states for the first letter of the surname, N for the first letter of the name and xxx are the last three digits of the identification number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (.fig e .m) have to be compressed (please select either the .zip or the .rar format) in a file named SNxxx.zip (o SNxxx.rar), which has to be uploaded to the beep portal of Politecnico di Milano. Each student is requested to:

1. Connect to the site <https://beep.metid.polimi.it>;
2. Enter the portal using its own identification code and password;
3. Select the section *Homework* of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select one of the following the folders *Test Dec. 11 2014 CS17- A*, *Test Dec. 11 2014 CS17- B*, *Test Dec. 11 2014 BL27- A*, *Test Dec. 11 2014 BL27- B*, according to the classroom and test version
5. Click on the button *Add* and select *Single document*;
6. Browse the compressed file SNxxx.zip using the button *Browse...* (or *Sfoggia...*)
7. Fill the Title field writing its own code SNxxx (e.g. for Bruni Stefano 123456: BS456);

Confirm the procedure clicking on the button *Publish*.